

Generation Interconnection Request System Impact Study Report

For Queue Project Q23-2331

Morony Dam 138 kV
June 02, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2331. The Interconnection Customer proposes a 100.8 MW solar photovoltaic facility interconnecting at the Morony Dam 138 kV substation in the New Jersey area. The request was received on July 07, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 677 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,429,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Magnolia 1708-Station E Toluca 138 kV	Crawfordsville Power Plant-Pine Tree Acres 138 kV	118.4	380	5.9
Sta E Touca-Locust Grove Substation 138 kV L	Pine Tree Acres Substation-Crawfordsville Power Plant 138 kV L	108.9	433	5.9
Pine Tree Acres-Linden Cogen Plant 138 kV	Linden Cogen Plant-Pine Tree Acres 138 kV #2	107.5	451	9.2
La Cygne-Proffit'S Spring 138 kV	Wisdom-Secord 138 kV Line	103.8	242	19.0
Thorndike Substation-Wisdom 138 kV line	La Cygne-Secord 138 kV	103.0	253	18.8
Thorndike-Crawfordsville Power Plant 138 kV	La Cygne-Secord 138 kV	102.5	224	18.8
Wisdom-Secord 138 kV L	Secord-La Cygne 138 kV Line	102.0	273	18.8
La Cygne-Secord 138 kV L	Secord-Wisdom 138 kV	101.3	291	19.0
Pine Tree Acres-Linden Cogen Plant 138 kV Ckt 2	Pine Tree Acres-Linden Cogen Plant 138 kV Cir. 1	100.6	482	9.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Magnolia 1708-Station E Toluca 138 kV L	Rebuild line with higher-capacity conductor	\$11,527,000	5% (\$531,000)
Sta E Touca-Locust Grove Substation 138 kV L	Reconductor line	\$411,000	5% (\$19,000)
Pine Tree Acres-Linden Cogen Plant 138 kV line	Reconductor line	\$663,000	9% (\$59,000)
Proffit'S Spring-La Cygne 138 kV	Reconductor line	\$379,000	15% (\$57,000)
Wisdom-Thorndike 138 kV line	Reconductor line	\$492,000	15% (\$74,000)
Thorndike-Crawfordsville Power Plant 138 kV	Reconductor line	\$1,794,000	15% (\$270,000)
Wisdom-Secord 138 kV L	Reconductor line	\$1,399,000	15% (\$210,000)
La Cygne-Secord 138 kV L	Reconductor line	\$1,072,000	15% (\$161,000)
Pine Tree Acres-Linden Cogen Plant 138 kV Ckt 2	Reconductor line	\$539,000	9% (\$48,000)
	Total allocated to project		\$1,429,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.